

CIRCUIT TOOL, DATABASES And FUN STUFF

The following software, besides other, can be found in the DVD accompanying EFY Plus edition of this magazine

MySQL 5.7

MySQL is a popular relational database management system written in C and C++. It is compatible with a long list of operating systems including Windows, MacOS, Linux, Solaris, FreeBSD, SunOS and Symbian. MySQL supports almost all major functionalities like stored procedure, triggers, cursors, views, SSL support, ACID compliance, subqueries, nested queries, unicode support and more.

Version 5.7 supports Replication function and binary-as-hex function to display binary data in hexadecimal format. It includes upgraded SSL security and bug fixes. In addition, the limitation of one trigger per action (in v. 5.6) has been removed. MySQL supports many usage interfaces including the main MySQL workbench, Adminer front-end interface for content management, DBedit database editor, LibreOffice Base and a graphical user interface for visual operations.

SQLite 3.21

If you are looking for a DBMS best suited for embedded applications, SQLite is the option for you. Instead of being a client-server-based engine, SQLite runs on the end application. It slightly differs from other proper DBMS. For example, instead of views, SQLite uses INSTEAD OF triggers. The variable type-assignment system is also different. Instead of a column-wise approach, it assigns types to variables individually.

The latest version includes support for JSON extension and a proper search engine. Some useful features of SQLite 3.21 include 8+3 Filename support, a backup API to copy content into local memory, memory-mapped input and output, null handling, WITHOUT ROW ID tables and write-ahead Log Mode.

WampServer 3.1

WampServer is a Windows web development software stack available for both 32-bit and 64-bit Windows operating systems. Its Version 3.1 comprises the latest version of Apache web browser, MySQL DBMS system, PHP programming and OpenSSL security. You can give online access to other users or restrict it to Localhost.

WampServer can also help you manage your server settings and access all logs. Installing WampServer is simple and well-instructed. In the main 'www' directory, you can create a sub-directory to place your PHP files.

MongoDB 3.6.0

MongoDB is a NoSQL Big Data database. This free and open source database uses JSON-like documents with schema as data source. MongoDB supports *ad hoc* queries like range queries, regular expression searches and user-defined JavaScript functions. It also includes replication, load balancing, aggregation and capped collection features. MongoDB can also be used as a file storage system.

The latest version of MongoDB has multiple bug fixes, improved Lookup function, new aggregator operations and variables, and improved security features.

Firebird 3.0

Firebird has gained immense popularity as an open source relational DBMS. This SQL-driven platform supports all major features of an ideal DBMS. These include stored procedures, triggers, user-defined functions and ACID-compliant transactions to name a few. It follows a multigenerational architecture that ensures locking mechanism does not affect smooth execution of OLTP and OLAP processes. Firebird has different access sources like ODBC, OLE DB, .NET, JDBC, API for native access, PHP or Perl.

Firebird 3.0 includes support for Boolean data-type, Data Definition Language (DDL) triggers, and Window functions in DML and IPV6 for connectivity.

VeeCAD

VeeCAD software converts your electronic circuit schematics into layouts. The schematic netlist (a combination of all the components, wires and patterns of a circuit) generated by a schematic design software is fed to VeeCAD, which uses the netlist to create a digital layout on stripboards.

VeeCAD comes with a strong components library. It also includes a variety of design tools including netlist import, component footprint definition, component positioning, connectivity check and writing.

Magarena 1.90

If you like card games, you will enjoy Magarena. It is an open source fantasy card game, where a single player plays against the computer as an opponent. Gameplay is based on the original card game *Magic: The Gathering*. Magarena is supported on Windows, Linux and MacOS.

The latest release, Magarena 1.90 has 15 new playable cards in addition to bug fixes and improved French translation.

DreamChess

Have fun playing chess on DreamChess. It is an open source game for Windows, MacOS and Linux platforms. The 3D OpenGL graphics greatly enhances the visuals. Players can also choose the chess board theme between classic wooden and flat figurines.

DreamChess has an AI called Dreamer against which players can play. Players can also integrate other X-Board compatible chess engines like GNU Chess or Crafty. DreamChess also adds effects like music and sound effects, movelist notation, savegames and undo move. **EFY**